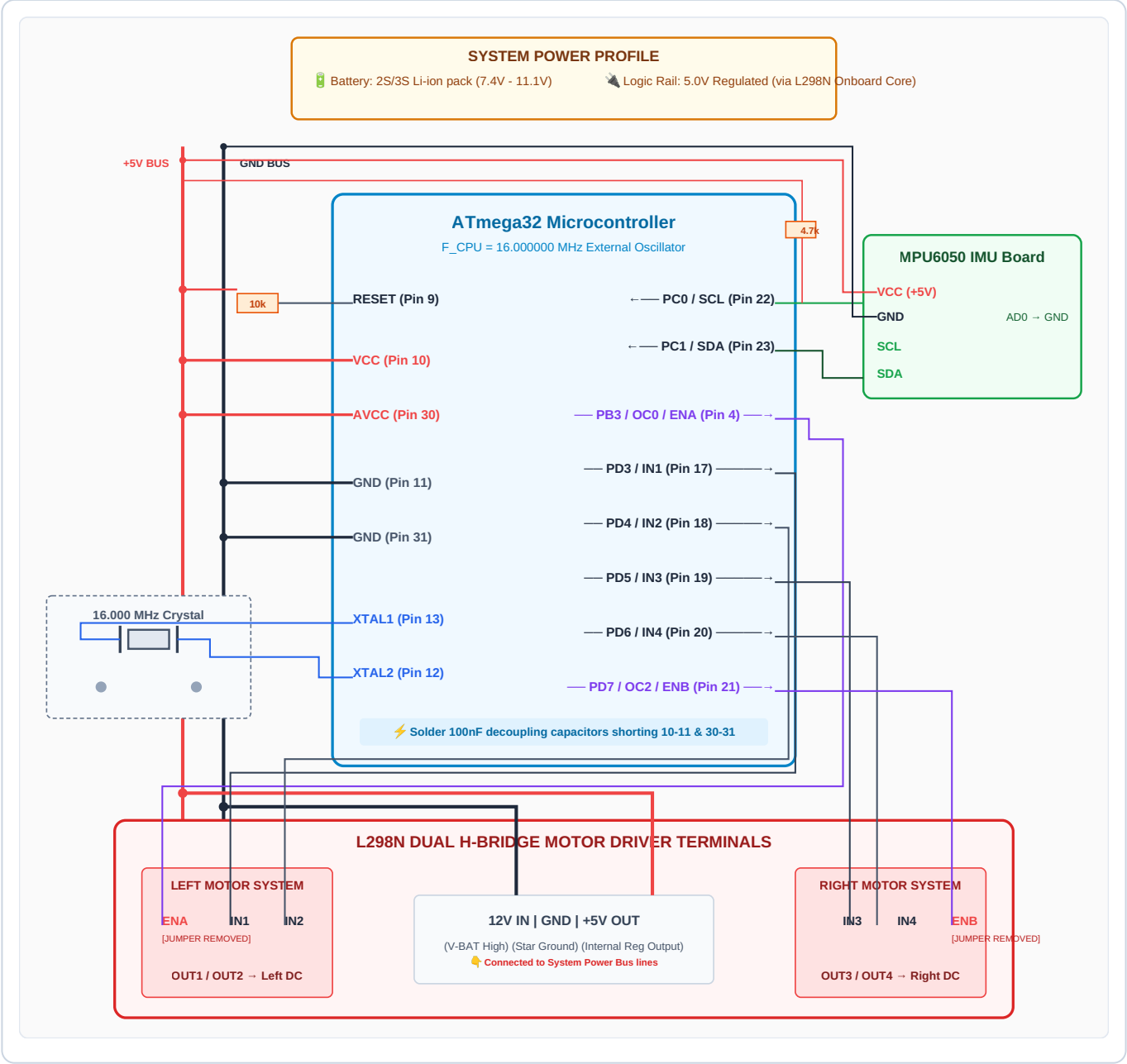


ATmega32 Self-Balancing Robot Master Schematic (V2)

⚡ **Fixed: Explicit VCC, AVCC, and Ground Paths Routed**

You are completely right. The previous version labeled the pins but left the actual power lines unrouted to keep the canvas clear. This updated diagram introduces explicit, color-coded physical connections for the **+5V Power Bus (Red Lines)** and the **Common Ground Bus (Black Lines)** running directly from the L298N regulator out to the microcontroller pins.



System Power Distribution Blueprint Matrix

Power Node Name	Physical Source Node	Target Component Destination Pins	Functional Wire Gauge & Installation Rule
+5V DC Logic Bus	L298N 5V OUT Screw Terminal	ATmega32 VCC (Pin 10) , AVCC (Pin 30) , MPU6050 VCC	Standard signal hookup wire. Bridges to Reset pin via 10kΩ pull-up.
Common Ground Bus	L298N GND Screw Terminal		

Power Node Name	Physical Source Node	Target Component Destination Pins	Functional Wire Gauge & Installation Rule
		ATmega32 GND (Pin 11) , GND (Pin 31) , MPU6050 GND & AD0	Star Ground Configuration: Converge all system ground return paths back to this point to isolate EMI.
High-Current Motor Rail	External Li-ion Battery pack (+)	L298N 12V IN Screw Terminal Input Node	Heavy gauge power line (20-22 AWG recommended) to avoid voltage drops during torque spikes.